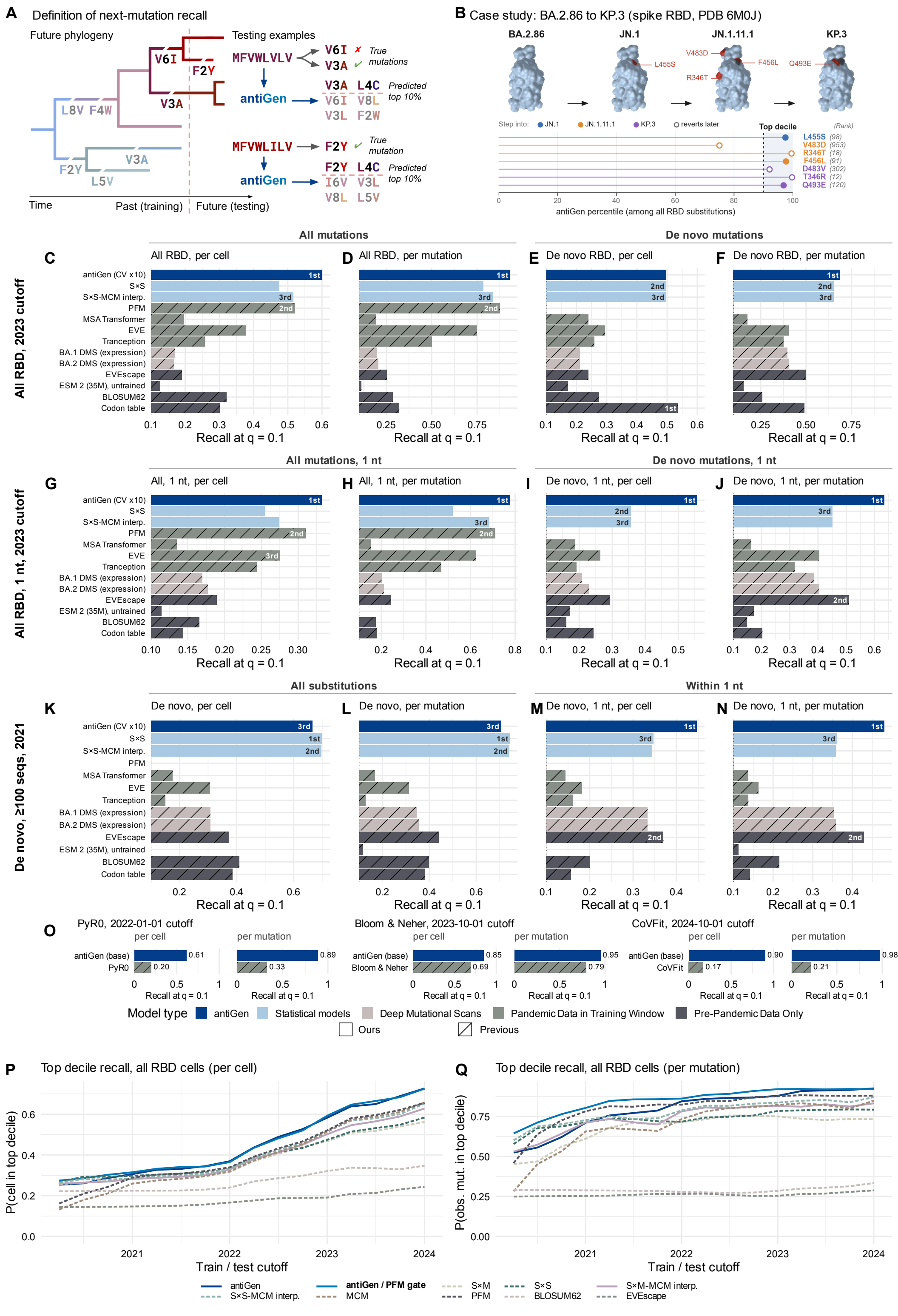

A
Definition of next-mutation recall
Future phylogeny
Testing examples
V6I
F2Y
V3A
L8V F4W
F2Y
V3A
L5V
Time
Past (training)
Future (testing)
MFVWLVLV
V6I ✗ True mutations
V3A ✓
antiGen
V3A L4C Predicted top 10%
V6I V8L
V3L F2W
MFVWLILV
F2Y ✓ True mutation
antiGen
F2Y L4C Predicted top 10%
I6V V3L
V8L L5V
B
Case study: BA.2.86 to KP.3 (spike RBD, PDB 6M0J)
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
F456L
R346T
Q493E
Step into: JN.1 JN.1.11.1 KP.3 reverts later
Top decile
(Rank)
L455S (98)
V483D (953)
R346T (18)
F456L (91)
D483V (302)
T346R (12)
Q493E (120)
0 20 40 60 80 100
antiGen percentile (among all RBD substitutions)
All mutations
De novo mutations
All RBD, 2023 cutoff
C
All RBD, per cell
D
All RBD, per mutation
E
De novo RBD, per cell
F
De novo RBD, per mutation
antiGen (CV x10)
S×S
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
BA.1 DMS (expression)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
BLOSUM62
Codon table
Recall at q = 0.1
All mutations, 1 nt
De novo mutations, 1 nt
All RBD, 1 nt, 2023 cutoff
G
All, 1 nt, per cell
H
All, 1 nt, per mutation
I
De novo, 1 nt, per cell
J
De novo, 1 nt, per mutation
All substitutions
Within 1 nt
De novo, ≥100 seqs, 2021
K
De novo, per cell
L
De novo, per mutation
M
De novo, 1 nt, per cell
N
De novo, 1 nt, per mutation
O
PyR0, 2022-01-01 cutoff
Bloom & Neher, 2023-10-01 cutoff
CoVFit, 2024-10-01 cutoff
per cell
per mutation
antiGen (base)
PyR0
0.61
0.20
0.89
0.33
Bloom & Neher
0.85
0.69
0.95
0.79
CoVFit
0.90
0.17
0.98
0.21
Model type
antiGen
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
P
Top decile recall, all RBD cells (per cell)
P(cell in top decile)
Train / test cutoff
Q
Top decile recall, all RBD cells (per mutation)
P(obs. mut. in top decile)
antiGen
S×S-MCM interp.
antiGen / PFM gate
MCM
S×M
PFM
S×S
BLOSUM62
S×M-MCM interp.
EVEscape